Lecture Slip, CS 235
Nov 3, 2026, Tuesday
Operators, Unique Strings, Trees

Name:	
EmpID:	

1. **Operator combinations.** The function `ops(v, k, expr)` from class fills every gap between the numbers in `v` with `+`, `-`, `*`, `/`.

   (a) How many lines does it print for `v = {1, 2, 3, 4}`?

   (b) After each recursive call, the code runs `expr.resize(len)`. Why not call `expr.pop_back()` twice instead?

2. **Printing unique strings.** `printUnique` keeps a `std::set<char> tried` inside each call, and runs `if (!tried.insert(s[i]).second) { continue; }`.

   (a) For `s = "ABB"` at the first call (`k = 0`), which values of `i` are skipped, and why?

   (b) How many lines do `printAll("ABB")` and `printUnique("ABB")` print?

3. **Make 24, Task B.** In Task B each value is a pair (text, number).

   (a) `arithmetic` is given `{"2", 2}` and `{"5", 5}`. Write the two **subtraction** entries it produces (text and value), in the required order $a - b$, then $b - a$.

   (b) In `make24`, a recursive call returns the empty string. What does the caller do next? What does the caller do when the string is *not* empty?

4. **Expression trees.** Consider $(6 - 2) * 3 + 1$.

   (a) Build the tree: write the operations in the order they are performed, then draw the tree.

   (b) Write the **pre-order** and the **post-order** traversals.

5. **Pace survey.** Please help me set the pace of the course. Circle one.

Too **slow** (I can go faster)	**Normal** (about right)	Too **fast** (please slow down)

If "too fast", which topic would you like me to revisit?

6. Do you have any questions about today's class? If so, please provide details. Thank you.